

EDEXCEL INTERNATIONAL A LEVEL

WME01/01 October 2019 Worked Solutions

October 2019 paper rebuilt with the worked solution after each question.

8

paper questions

WME01

Mechanics 1

**Single-
paper
solutions**standalone IAL
route**Dr Eslam Ahmed**Prepared for Dr Eslam Ahmed - eliteigcse.com**M1**

PAST PAPER

WME01/01 October 2019

October 2019 | 8 questions | 75 marks

8

questions

75

marks

Answers

worked solution
after each
question

Question 1

Momentum, Impulse & Collisions

1. Two particles, P and Q , have masses $3m$ and $2m$ respectively. The particles are connected by a light inextensible string. Initially P and Q are at rest on a smooth horizontal plane with the string slack.

Particle P is then projected along the plane directly away from Q with speed $4u$. At the same instant, particle Q is projected along the plane in the opposite direction with speed $3u$.

Find

- (a) the common speed of the particles immediately after the string becomes taut, (3)
- (b) the magnitude of the impulse exerted on Q at the instant when the string becomes taut. (3)

Worked Solution - Question 1

Topic group

1. Choose a direction

Take P's initial direction as positive. Then P initially has velocity $4u$ and Q initially has velocity $-3u$.

2. Use conservation of momentum

When the string becomes taut, both particles move with common speed v . Hence $3m(4u) + 2m(-3u) = (3m + 2m)v$.

3. Find the common speed

$$12mu - 6mu = 5mv, \text{ so } v = \frac{6u}{5}.$$

4. Use impulse on Q

$$\text{Impulse on Q equals change in momentum: } I = 2m \left(\frac{6u}{5} - (-3u) \right).$$

5. Find the impulse magnitude

$$I = 2m \left(\frac{6u}{5} + \frac{15u}{5} \right) = 2m \cdot \frac{21u}{5} = \frac{42}{5}mu.$$

Final answer

$$(a) v = \frac{6u}{5}. \quad (b) I = \frac{42}{5}mu.$$

Question 2

Constant Acceleration in 1D

2. A small ball is released from rest from a point that is 40 m above horizontal ground. The ball bounces on the ground and rebounds vertically. Each time the ball bounces on the ground, the speed of the ball is instantaneously reduced by 50%. The ball is modelled as a particle moving freely under gravity, from the instant when it is released until it first hits the ground, and between each successive bounce.
- (a) Find the time from the instant when the ball is released from rest to the instant when it hits the ground for the second time. (5)
- (b) Find the maximum height reached by the ball above the ground after the ball's third bounce. (4)

Worked Solution - Question 2

Topic group

1. Find the time to the first hit

The ball is released from rest and falls 40 m. Use $s = ut + \frac{1}{2}gt^2$: $40 = \frac{1}{2}gt_1^2$.

$$\text{Thus } t_1 = \sqrt{\frac{80}{g}} = \frac{20}{7} \text{ s.}$$

2. Find the speed just before the first bounce

Using $v^2 = u^2 + 2gs$, $v_1^2 = 2g(40)$, so $v_1 = 28 \text{ m s}^{-1}$.

3. Use the rebound speed

After the first bounce the speed is reduced by 50%, so the upward speed is 14 m s^{-1} .

4. Find the time until the second hit

The time to go up and return to the ground is $\frac{2(14)}{g} = \frac{20}{7} \text{ s}$.

5. Find the total time

Total time to the second hit is $\frac{20}{7} + \frac{20}{7} = \frac{40}{7} = 5.71 \text{ s}$ approximately.

6. Find the speed after the third bounce

After each bounce, the speed is halved. The upward speeds after the first, second and third bounces are 14, 7 and 3.5 m s^{-1} .

7. Find the height after the third bounce

At the top after the third bounce, $0 = (3.5)^2 - 2gh$. Therefore

$$h = \frac{(3.5)^2}{2g} = 0.625 \text{ m.}$$

Final answer

(*a*) 5.71 s approximately. (*b*) 0.625 m.

Question 3

Newton's Second Law

3. A car of mass 800 kg is towing a trailer of mass 400 kg up a straight road using a towbar. The towbar is parallel to the road and parallel to the direction of motion of the car. The road is inclined to the horizontal at an angle α , where $\sin \alpha = \frac{1}{7}$. The engine of the car produces a constant driving force of magnitude D newtons. The resistance to the motion of the car from non-gravitational forces is modelled as a single force of magnitude 420 N. The resistance to the motion of the trailer from non-gravitational forces is modelled as a single force of magnitude 300 N. The car and trailer are modelled as particles and the towbar is modelled as a light rod.

Given that the tension in the towbar is 2060 N, find the value of D .

(7)

Worked Solution - Question 3

1. Use the trailer to find acceleration

For the trailer, take up the road as positive. The forces along the road are tension **2060** N up the road, resistance **300** N down the road, and weight component $400g \sin \alpha$ down the road.

2. Form the trailer equation

$2060 - 300 - 400g \sin \alpha = 400a$. Since $\sin \alpha = \frac{1}{7}$, this becomes
 $2060 - 300 - 400g \cdot \frac{1}{7} = 400a$.

3. Find the acceleration

Using $g = 9.8$, $400g \cdot \frac{1}{7} = 560$. Hence $1200 = 400a$ and $a = 3 \text{ m s}^{-2}$.

4. Apply Newton's second law to the car

For the car: $D - 420 - 800g \sin \alpha - 2060 = 800a$.

5. Find the driving force

$D - 420 - 800g \cdot \frac{1}{7} - 2060 = 800(3)$. Since $800g \cdot \frac{1}{7} = 1120$,
 $D = 420 + 1120 + 2060 + 2400 = 6000 \text{ N}$.

Final answer

$$D = 6000 \text{ N}.$$

Question 4

Resolving Forces, Inclined Planes

4.

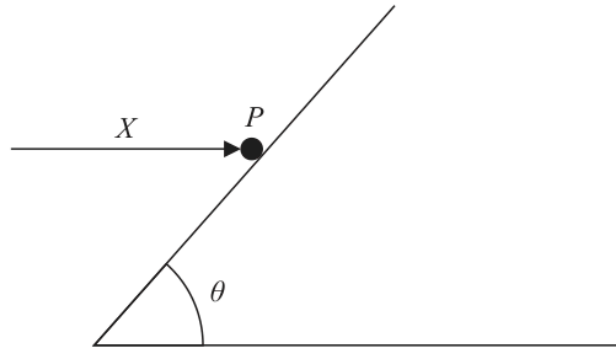


Figure 1

A particle, P , of mass km lies on a fixed rough plane. The plane is inclined to the horizontal at an acute angle θ . A horizontal force of magnitude X acts on P , as shown in Figure 1. The line of action of the force lies in the vertical plane which contains the line of greatest slope of the inclined plane that passes through P . The coefficient of friction between P and the inclined plane is μ .

When $X = mg$, the particle P is in equilibrium and on the point of sliding down the plane.

(a) Show that $\mu = \frac{k \tan \theta - 1}{k + \tan \theta}$ (10)

(b) Deduce that, when $k = 1$, θ must be greater than 45° (2)

Worked Solution - Question 4

1. Set friction direction

The particle is on the point of sliding down the plane, so friction acts up the plane. Also $X = mg$.

2. Resolve perpendicular to the plane

Both the weight kmg and the horizontal force mg press the particle into the plane. Therefore $R = kmg \cos \theta + mg \sin \theta$.

3. Resolve parallel to the plane

Down the plane is the component $kmg \sin \theta$. Up the plane are friction F and the component $mg \cos \theta$. Thus $F + mg \cos \theta = kmg \sin \theta$.

4. Write friction in terms of theta

So $F = kmg \sin \theta - mg \cos \theta$.

5. Use limiting friction

$$\mu = \frac{F}{R} = \frac{kmg \sin \theta - mg \cos \theta}{kmg \cos \theta + mg \sin \theta}.$$

6. Simplify the expression

Cancel mg and divide numerator and denominator by $\cos \theta$: $\mu = \frac{k \tan \theta - 1}{k + \tan \theta}$, as required.

7. Deduce the condition when k is 1

When $k = 1$, $\mu = \frac{\tan \theta - 1}{1 + \tan \theta}$. Since $\mu > 0$ and the denominator is positive, $\tan \theta - 1 > 0$.

8. Finish the inequality

Therefore $\tan \theta > 1$, so because θ is acute, $\theta > 45^\circ$.

Final answer

$$(a) \mu = \frac{k \tan \theta - 1}{k + \tan \theta}. \quad (b) \theta > 45^\circ.$$

Question 5

Moments

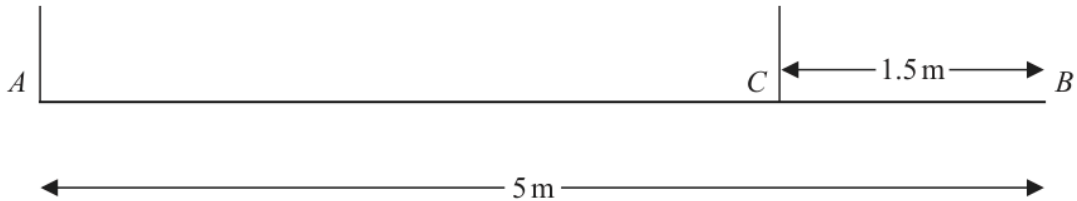


Figure 2

A non-uniform beam, AB , has length 5 m and mass 12 kg. The beam is suspended in a horizontal position by two vertical ropes. One rope is attached to the beam at A . The other rope is attached to the beam at C , where $CB = 1.5$ m, as shown in Figure 2. The distance of the centre of mass of the beam from A is 1.75 m. The beam is modelled as a non-uniform rod and the ropes are modelled as light inextensible strings.

A particle of mass M kg is now placed on the beam at B and the beam remains in equilibrium in a horizontal position.

- (a) Find the largest possible value of M . (3)

The particle at B is now removed and a particle of mass 15 kg is now placed on the beam at the point D , where $AD = x$ metres. The beam remains in equilibrium in a horizontal position.

Given that the tension in the rope attached to the beam at C is now twice the tension in the rope attached to the beam at A ,

- (b) find the value of x . (5)

Worked Solution - Question 5

1. Locate the key distances

The beam is 5 m long. Since $CB = 1.5$ m, $AC = 3.5$ m. The centre of mass is 1.75 m from A, so it is also 1.75 m from C.

2. Use the largest M condition

For the largest possible M at B, the rope at A is just slack, so the beam is supported at C only.

3. Take moments about C

The particle at B is 1.5 m to the right of C, and the beam's weight acts 1.75 m to the left of C. Hence $Mg(1.5) = 12g(1.75)$.

4. Find M

$$M = \frac{12(1.75)}{1.5} = 14.$$

5. Set the tensions for part b

Let the tension at A be T . The tension at C is $2T$. Vertical equilibrium gives $T + 2T = 12g + 15g = 27g$, so $T = 9g$ and the tension at C is $18g$.

6. Take moments about A

For the new particle at $AD = x$, moments about A give $18g(3.5) = 12g(1.75) + 15gx$.

7. Solve for x

Cancel g : $63 = 21 + 15x$. Thus $15x = 42$ and $x = 2.8$ m.

Final answer

(a) $M = 14$. (b) $x = 2.8$ m.

Question 6

Kinematics Graphs

6. An athlete runs a 200 m race along a straight horizontal track.

In a model of the motion of the athlete, air resistance is ignored, the athlete starts from rest at time $t = 0$ seconds and moves with uniform acceleration 0.8 m s^{-2} for T seconds, reaching a speed of $V \text{ m s}^{-1}$. She then maintains this speed until she crosses the finishing line.

The total time from when the athlete starts to when she crosses the finishing line is 30 s.

- (a) Sketch a speed-time graph for the model of the motion of the athlete from the instant when she starts to the instant when she crosses the finishing line. (2)
- (b) Write down an expression for V in terms of T . (1)
- (c) Show that $T^2 - kT + 500 = 0$, where k is a constant to be found. (4)
- (d) Hence find the value of T , justifying your answer carefully. (3)
- (e) Considering your speed-time graph or otherwise, state two ways, apart from including air resistance, in which the model could be made to be more realistic. (2)

Worked Solution - Question 6

1. Describe the speed-time graph

The graph starts at $(0, 0)$, rises in a straight line with gradient 0.8 until time T and speed V , then stays horizontal at speed V until $t = 30$.

2. Write V in terms of T

Uniform acceleration gives $V = 0.8T$.

3. Use area for the distance

The distance is the area under the speed-time graph: $\frac{1}{2}TV + V(30 - T) = 200$.

4. Substitute V

Using $V = 0.8T$: $\frac{1}{2}T(0.8T) + (0.8T)(30 - T) = 200$.

5. Form the quadratic

$0.4T^2 + 24T - 0.8T^2 = 200$, so $-0.4T^2 + 24T - 200 = 0$. Multiplying by -2.5 gives $T^2 - 60T + 500 = 0$, so $k = 60$.

6. Solve and choose the valid root

$T^2 - 60T + 500 = (T - 10)(T - 50) = 0$, so $T = 10$ or 50 . Since the whole race lasts 30 s, $T = 10$.

7. Give two model improvements

A more realistic model could include reaction time at the start, and could replace the instantaneous change from acceleration to constant speed with a smoother transition.

Final answer

(a) Speed-time graph rises linearly to V at $t = T$, then is horizontal until $t = 30$.

(b) $V = 0.8T$. (c) $T^2 - 60T + 500 = 0$, so $k = 60$.

(d) $T = 10$. (e) For example, include reaction time at the start and avoid the instantaneous change to constant speed.

Question 7

Working with Vectors

7. Two forces, $\mathbf{F}$ and $\mathbf{G}$, act on a particle. The force $\mathbf{F}$ has magnitude 4N and acts in a direction with a bearing of 120° and the force $\mathbf{G}$ has magnitude 6N and acts due north.

Given that $\mathbf{P} = 2\mathbf{F} + \mathbf{G}$, find

- (i) the magnitude of $\mathbf{P}$
- (ii) the direction of $\mathbf{P}$, giving your answer as a bearing to the nearest degree.

(7)

Worked Solution - Question 7

1. Resolve $2\mathbf{F}$

Since $\mathbf{F}$ has magnitude 4 N on a bearing of 120° , $2\mathbf{F}$ has magnitude 8 N on the same bearing.

2. Write the components

$$2\mathbf{F} = 8 \sin 120^\circ \mathbf{i} + 8 \cos 120^\circ \mathbf{j} = 4\sqrt{3}\mathbf{i} - 4\mathbf{j}. \text{ Also } \mathbf{G} = 6\mathbf{j}.$$

3. Find $\mathbf{P}$

$$\mathbf{P} = 2\mathbf{F} + \mathbf{G} = 4\sqrt{3}\mathbf{i} + 2\mathbf{j}.$$

4. Find the magnitude

$$|\mathbf{P}| = \sqrt{(4\sqrt{3})^2 + 2^2} = \sqrt{48 + 4} = \sqrt{52} \text{ N}.$$

5. Find the bearing angle

The vector is in the north-east quadrant. The angle east of north satisfies

$$\tan \theta = \frac{4\sqrt{3}}{2} = 2\sqrt{3}.$$

6. State the bearing

$\theta = 73.9^\circ$ approximately, so the bearing is 074° to the nearest degree.

Final answer

(i) $|\mathbf{P}| = \sqrt{52} \text{ N} \approx 7.21 \text{ N}$. (ii) bearing = 074° .

Question 8

Working with Vectors

8. [In this question, the horizontal unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed due east and due north respectively and position vectors are given relative to a fixed origin O .]

Two speedboats, A and B , are each moving with constant velocity. The velocity of A is 20 km h^{-1} due west and the velocity of B is 40 km h^{-1} on a bearing of 150° . The boats are modelled as particles.

At noon, the position vector of A is $60\mathbf{i} \text{ km}$ and B is at the origin O . At time t hours after noon, the position vector of A is $\mathbf{r} \text{ km}$ and the position vector of B is $\mathbf{s} \text{ km}$.

- (a) Find the velocity of B in the form $(p\mathbf{i} + q\mathbf{j}) \text{ km h}^{-1}$ (3)
- (b) Find expressions for $\mathbf{r}$ and $\mathbf{s}$ in terms of t , $\mathbf{i}$ and $\mathbf{j}$. (3)
- (c) Find the time, to the nearest minute, at which the distance between the boats is the same as it was at noon. (8)

Worked Solution - Question 8

1. Resolve B's velocity

A bearing of 150° is 30° east of south. Therefore the east component is $40 \sin 150^\circ = 20$ and the north component is $40 \cos 150^\circ = -20\sqrt{3}$.

2. Write B's velocity

$$\mathbf{v}_B = 20\mathbf{i} - 20\sqrt{3}\mathbf{j} \text{ km h}^{-1}.$$

3. Write A's position

Boat A starts at $60\mathbf{i}$ and travels due west at 20 km h^{-1} , so $\mathbf{r} = 60\mathbf{i} - 20t\mathbf{i} = (60 - 20t)\mathbf{i}$.

4. Write B's position

Boat B starts at O, so $\mathbf{s} = t(20\mathbf{i} - 20\sqrt{3}\mathbf{j}) = 20t\mathbf{i} - 20\sqrt{3}t\mathbf{j}$.

5. Find the separation vector

$$\overrightarrow{AB} = \mathbf{s} - \mathbf{r} = (20t - (60 - 20t))\mathbf{i} - 20\sqrt{3}t\mathbf{j} = (40t - 60)\mathbf{i} - 20\sqrt{3}t\mathbf{j}.$$

6. Use the noon distance

At noon the distance between the boats is 60 km. Set the later distance equal to 60: $(40t - 60)^2 + (-20\sqrt{3}t)^2 = 60^2$.

7. Solve for t

Expanding gives $2800t^2 - 4800t = 0$, so $t(2800t - 4800) = 0$. The later time is $t = \frac{4800}{2800} = \frac{12}{7}$ hours.

8. Convert to clock time

$\frac{12}{7}$ hours is 1 hour and 42.857... minutes after noon, which rounds to **1:43** pm, or **13:43**.

Final answer

(a) $\mathbf{v}_B = 20\mathbf{i} - 20\sqrt{3}\mathbf{j} \text{ km h}^{-1}$. (b) $\mathbf{r} = (60 - 20t)\mathbf{i}$, $\mathbf{s} = 20t\mathbf{i} - 20\sqrt{3}t\mathbf{j}$. (c) 13:43, or 1:43 pm.